

<i>Title:</i>	Revision of the Bachelor of Science in Chemical Engineering
<i>Sponsor:</i>	<i>Department of Chemical Engineering</i>
<i>Executive Summary:</i>	The department is proposing to add a new technical elective course, CHE 453 Fundamentals of Electrochemistry to the chemical engineering program. The course was developed under the general CHE 494 "Selective Topics in Chemical Engineering" rubric and has been offered 3 times (<i>Spring 2014, Spring 2017, Spring 2019</i>). It will be offered to undergraduate students as a 3 credit hour course and to graduate students as a 4 credit hour course and will give the students another course option to fulfill their technical elective requirement.
<i>Description:</i>	Addition of one technical elective inside the major rubric; new course (CHE 453). This change does not affect the total credit hours required for the program.
<i>Justification:</i>	The course provides graduate students and senior undergraduate students the theory, principles and the latest advances in electrochemistry. There has been renewed interest in the subject of electrochemistry due to the recent advances in energy storage technology, and this class will provide a foundation for students interested in working in this area. The course has been well-received by undergraduate and graduate chemical engineering students and satisfies the technical elective requirement for the bachelor's degree. The topics covered in the course also integrate holistically with the overall Chemical Engineering curriculum and undergraduate student learning experience.
<i>Catalog Statement:</i>	<i>See below</i>
<i>Minority Impact Statement:</i>	<i>The changes will clarify the degree requirements for all students and will not adversely impact minority students.</i>
<i>Budgetary and Staff Implications:</i>	<i>None expected</i>
<i>Library Resource Implications:</i>	<i>None expected</i>

<i>Space Implications:</i>	<i>None expected</i>
<i>Unit (e.g. department) approval date:</i> <i>College (educational policy committee, faculty) approval dates:</i>	Provide unit dates, including department, college/school committees, faculty Approvals, as applicable. Chemical engineering faculty approval: 01/23/19 Chemical Engineering department approval: 09/13/19 Education Policy Committee approval:
<i>Contact Person:</i>	Alan Zdunek zdunek@uic.edu
<i>Proposed Effective Date/Term:</i>	Spring 2020

Current	Revised
BS in Chemical Engineering	BS in Chemical Engineering
<i>Degree Requirements</i> To earn a Bachelor of Science in Chemical Engineering degree from UIC, students need to complete University, college, and department degree requirements. The Department of Chemical Engineering degree requirements are outlined below. Students should consult the <i>College of Engineering</i> section for additional degree requirements and college academic policies.	SAME
BS in Chemical Engineering Degree Requirements Hours Nonengineering and General Education Requirements 73 Required in the College of Engineering 49 Technical Elective ^a 3 Electives outside the Major Rubric ^a 3 Total Hours—BS in Chemical Engineering 128	SAME
<i>a Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.</i>	
Nonengineering and General Education Requirements ENGL 160— Academic Writing I: Writing for Academic and Public Contexts 3 ENGL 161— Academic Writing II: Writing for Inquiry and Research 3 Exploring World Cultures course ^a 3 Understanding the Creative Arts course ^a 3 Understanding the Past course ^a 3 Understanding the Individual and Society course ^a 3 Understanding U.S. Society course ^a 3 MATH 180—Calculus I ^b 4	SAME

MATH 181—Calculus II ^b	4	
MATH 210—Calculus III ^b	3	
MATH 220—Introduction to Differential Equations	3	
PHYS 141—General Physics I (Mechanics) ^b	4	
PHYS 142—General Physics II (Electricity and Magnetism) ^b	4	
CHEM 122- General College Chemistry I ^b (Lecture)	4	
CHEM 123- General College Chemistry I ^b (Laboratory)	1	
<u>OR</u> CHEM 116 – Honors and Major General and Analytical Chemistry I	5	
CHEM 124—General College Chemistry II ^b (Lecture)	4	
CHEM 125—General College Chemistry II ^b (Lab)	1	
<u>OR</u> CHEM 118 – Honors and Major General and Analytical Chemistry II	5	
CHEM 222—Analytical Chemistry ^c	4	
CHEM 232—Organic Chemistry I	4	
CHEM 233- Organic Chemistry Laboratory	2	
CHEM 234—Organic Chemistry II	4	
CHEM 342—Physical Chemistry I	3	
CHEM 346—Physical Chemistry II	3	
Total Hours- Nonengineering and General Education Requirements	73	
a Students should consult the General Education section of the catalog for a list of approved courses in this category. b This course is approved for the Analyzing the Natural World General Education category. ^c Students who take CHEM 116 and CHEM 118 to fulfill the general chemistry requirement do not need to take CHEM 222. Instead they should enroll in one of the following: CHEM 314: Inorganic Chemistry; CHEM 452: Biochemistry; or CHEM 402: Chemical Information Systems and CHEM 444: Physical Chemistry III.		
Required in the College of Engineering		SAME
Courses	Hours	
ENGR 100—Orientation ^a	0 ^a	
CHE 201—Introduction to Thermodynamics	3	
CHE 205—Computational Methods in Chem. Eng.	3	
CHE 210—Material and Energy Balances	4	
CHE 301—Chemical Engineering: Thermodynamics	3	
CHE 311—Transport Phenomena I	3	
CHE 312—Transport Phenomena II	3	
CHE 313—Transport Phenomena III	3	
CHE 321—Chemical Reaction Engineering	3	
CHE 341—Chemical Process Control	3	
CHE 381—Chemical Engineering Laboratory I	2	
CHE 382—Chemical Engineering Laboratory II	2	
CHE 396—Senior Design I	4	
CHE 397—Senior Design II	4	
CHE 499—Professional Development Seminar	0	
ECE 210—Electrical Circuit Analysis	3	
CME 260—Properties of Materials	3	
CS 109—C/C++ Programming for Engineers with MatLab	3	

Total Hours—Required in the College of Engineering		49
a ENGR 100 is one-semester-hour course, but the hour does not count toward the total hours required for graduation.		
Technical Elective		
Courses	Hours	Courses
<i>One technical elective to be chosen from the following list of design-oriented courses:^a</i>		<i>One technical elective to be chosen from the following list of design-oriented courses:^a</i>
CHE 392—Undergraduate Research (3) ^b	3	CHE 392—Undergraduate Research (3) ^b
CHE 410—Transport Phenomena (3)		CHE 410—Transport Phenomena (3)
CHE 413—Introduction to Flow in Porous Media (3)		CHE 413—Introduction to Flow in Porous Media (3)
CHE 421—Combustion Engineering (3)		CHE 421—Combustion Engineering (3)
CHE 422—Biochemical Engineering (3)		CHE 422—Biochemical Engineering (3)
CHE 423—Catalytic Reaction Engineering (3)		CHE 423—Catalytic Reaction Engineering (3)
CHE 425 Nanotechnology Drug Deliver (3)		CHE 425 Nanotechnology Drug Deliver (3)
CHE 431—Numerical Methods in Chemical Engineering (3)		CHE 431—Numerical Methods in Chemical Engineering (3)
CHE 433 Process Simulation with Aspen Plus (3)		CHE 433 Process Simulation with Aspen Plus (3)
CHE 438—Computational Molecular Modeling (3)		CHE 438—Computational Molecular Modeling (3)
CHE 440—Non-Newtonian Fluids (3)		CHE 440—Non-Newtonian Fluids (3)
CHE 441—Computer Applications in Chemical Engineering (3)		CHE 441—Computer Applications in Chemical Engineering (3)
CHE 445—Mathematical Methods in Chemical Engineering (3)		CHE 445—Mathematical Methods in Chemical Engineering (3)
CHE 450—Air Pollution Engineering (4)		CHE 450—Air Pollution Engineering (4)
CHE 451 Renewable Energy Technologies (3)		CHE 453- Fundamentals of Electrochemistry (3)
CHE 456—Fundamentals and Design of Microelectronics Processes 930		CHE 456—Fundamentals and Design of Microelectronics Processes 930
CHE 494—Selected Topics in Chemical Engineering (3)		CHE 494—Selected Topics in Chemical Engineering (3)
Total Hours—Technical Elective	3	Total Hours—Technical Elective
a Possible technical elective credit for a 400-level CHE course not listed above will require departmental approval by petition to the Undergraduate Committee.		
b An appropriate design-related research project may be selected with the approval of the Department of Chemical Engineering.		
Courses	Hours	SAME
Electives outside the CHE rubric	3	
Total Hours—Electives outside the Major Rubric	3	SAME
BS in Chemical Engineering, Bioengineering Concentration		
Students in this concentration complete the following:		
Required Courses ^a		
Technical Elective		
<u>CHE 422</u>	Biochemical Engineering	3

Electives	
Select two electives in nonmajor rubric category from the following:	5-7
<u>BIOS 350</u> General Microbiology	
<u>BIOS 351</u> Microbiology Laboratory	
<u>CHEM 352</u> Introductory Biochemistry	
<u>CHEM 452</u> Biochemistry I	
Total Hours	8-10
^a Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 130 semester hours for the degree.	
Sample Course Schedule	
Freshman Year	
First Semester	Hours
MATH 180—Calculus I	4
CHEM 122—General College Chemistry I (Lecture)	4
CHEM 123—General College Chemistry I (Lab)	4
OR CHEM 116 – Honors and Major General and Analytical Chemistry I	5
ENGL 160—Academic Writing I: Writing for Academic and Public Contexts	3
CS 109—C/C++ Programming for Engineers with MatLab	3
ENGR 100—Orientation ^a	1 ^a
Total Hours	15
Second Semester	Hours
MATH 181—Calculus II	4
PHYS 141—General Physics I (Mechanics)	4
ENGL 161—Academic Writing II: Writing for Inquiry and Research	3
CHEM 124—General College Chemistry II (Lecture)	4
CHEM 125—General College Chemistry II (Lab)	1
OR CHEM 118– Honors and Major General and Analytical Chemistry II	5
Total Hours	16
Sophomore Year	
First Semester	Hours
MATH 210—Calculus III	3
PHYS 142—General Physics II (Electricity and Magnetism)	4
CHEM 232—Organic Chemistry I	4
CHE 201—Introduction to Thermodynamics	3
General Education Core course	3

SAME
SAME

Total Hours	17	
Second Semester	Hours	
MATH 220—Introduction to Differential Equations	3	
CHEM 234—Organic Chemistry II	4	SAME
CHEM 233—Organic Chemistry Lab I	2	
CHE 205—Computational Methods in Chem. Eng.	3	
CHE 210—Material and Energy Balances	4	
CME 260— Properties of Materials	3	
Total Hours	19	
Junior Year		SAME
First Semester	Hours	
ECE 210—Electrical Circuit Analysis	3	
CHE 301—ChE Thermodynamics	3	
CHE 311—Transport Phenomena I	3	
CHEM 222—Analytical Chemistry	4	
General Education Core course	3	
Total Hours	16	
Second Semester	Hours	
CHEM 342—Physical Chemistry I	3	
CHE 312—Transport Phenomena II	3	
CHE 313— Transport Phenomena III	3	
CHE 321—Chemical Reaction Engineering	3	
General Education Core course	3	
Total Hours	15	
Senior Year		
First Semester	Hours	
CHE 381—Chemical Engineering Laboratory I	2	Senior Year
CHE 396—Senior Design I	4	First Semester
CHEM 346—Physical Chemistry II	3	Hours
CHE Technical elective—Selected from CHE 410, 413, 421, 422, 423, 431, 438, 440, 441, 445, 450, 456, 494, or 392 (departmental approval is required for CHE 392)	3	CHE 381—Chemical Engineering Laboratory I
General Education Core Course	3	2
Total Hours	15	CHE 396—Senior Design I
		4
		CHEM 346—Physical Chemistry II
		3
		CHE Technical elective—Selected from CHE 410, 413, 421, 422, 423, 431, 438, 440, 441, 445, 450, 451 , 456, 494, or 392 (departmental approval is required for CHE 392)
		3
		General Education Core Course
		3
		Total Hours
		15
Second Semester	Hours	
CHE 382—Chemical Engineering Laboratory II	2	
CHE 341—Chemical Process Control	3	Same
CHE 397—Senior Design II	4	
CHE 499—Professional Development Seminar	0	
Elective outside the major rubric	3	
General Education Core course	3	
Total Hours	15	

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